

What is Happening to Thermography Tools



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What is a quick and effective way to identify defects on a circuit board assembly without conventional troubleshooting? The answer is thermography. Thermal infrared (IR) imaging devices are excellent tools for increasing the life expectancy of devices by getting real-time imagery of temperature distribution as well as development of heat on objects in a board through a contact-less tool.

Nip it in the bud

Higher-resolution thermal imaging cameras can spot overheating in individual components like transistors and resistors across the whole printed circuit board (PCB). This means that engineers can use it in an early-stage design to sort out circuitry damages. For example, a component that is not designed to go above 85°C could end up above 110°C and get stressed by overheating. The thermal camera immediately catches the over-heated component in its viewfinder without the engineer having to probe each component individually. This means that chances for an over-cooked component to ruin a PCB are far lesser since the engineer can see almost the entire board at a glance.

Earlier this year, Keysight had announced higher-temperature models, targeting electrical and electronic applications where these can be used to detect variations over a wider temperature range. U5856A and U5857A can perform temperature measurements up to 650°C and 1200°C, respectively.

Fluke and FLIR also launched similarly-named models:

Fluke's TiX1000 and FLIR's T1K or T1020. Both these models have very good

What goes inside newer thermal imagers

FLIR's T1K series of thermal imagers come with a full-high-definition (FHD) sensor paired with HDIR lenses.

Fluke's TiX series of thermal imagers feature 100 per cent diamond-turned-germanium lenses covered with a special coating that is designed to transmit energy to the detector with very good efficiency.

FLIR also released Muon core for thermal imaging, which simplifies the integration of thermal imaging capabilities for original equipment manufacturers. You might find it inside upcoming thermal imagers from other firms, too.

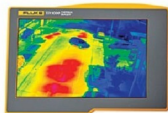
Thermal image overlaying CMOS images technology

Most innovative advances in thermal imagers for electronics engineers over the past few years include thermal image overlaying complementary metal-oxide semiconductor (CMOS) image technology. It offers a better thermal and image resolution for object identification to locate the bug from thermal gradient of circuitry, easily and quickly.

We are able to offer a wider industrial temperature range of thermal cameras to bridge the original industrial IR thermometer user and commercial thermal camera user with a better resolution of thermal cameras with CMOS overlaying graphic features and precise distance-to-spot ratio laser in circle indication.

—David Ko, general manager, Flex Instruments Co. Ltd

Fluke TiX1000



specifications; details of these have been included in the table.

Upgrades for displays and sensors

Fluke's new expert series thermal imagers called TiX560 and TiX520 come with an articulating lens that can rotate 180° as well as a large touchscreen LCD featuring a resolution of 320 × 240. The camera also